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Organ doses can be estimated from the computed tomography (CT) dose index for cone-beam CT on radiotherapy equipment

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Abstract

Cone beam computed tomography (CBCT) systems are fitted to radiotherapy linear accelerators and used for patient positioning prior to treatment by image guided radiotherapy (IGRT). Radiotherapists' and radiographers' knowledge of doses to organs from CBCT imaging is limited. The weighted CT dose index for a reference beam of width 20 mm ($CTDI_{w,ref}$) is displayed on Varian CBCT imaging equipment known as an On-Board Imager (OBI) linked to the Truebeam linear accelerator. This has the potential to provide an indication of organ doses. This knowledge would be helpful for guidance of radiotherapy clinicians preparing treatments. Monte Carlo simulations of imaging protocols for head, thorax and pelvic scans have been performed using EGSnrc/BEAMnrc, EGSnrc/DOSXYZnrc, and ICRP reference computational male and female phantoms to derive the mean absorbed doses to organs and tissues, which have been compared with values for the $CTDI_{w,ref}$ displayed on the CBCT scanner console. Substantial variations in dose were observed between male and female phantoms. Nevertheless, the $CTDI_{w,ref}$ gave doses within $\pm 21\%$ for the stomach and liver in thorax scans and $2 \times CTDI_{w,ref}$ can be used as a measure of doses to breast, lung and oesophagus. The $CTDI_{w,ref}$ could provide indications of doses to the brain for head scans, and the colon for pelvic scans. It is proposed that knowledge of the link between $CTDI_w$ for CBCT should be promoted and included in the training of radiotherapy staff.

Keywords: cone beam computed tomography, computed tomography dose index, organ dose, Monte Carlo dose calculation, ICRP reference phantom, image guided radiotherapy

(Some figures may appear in colour only in the online journal)

1. Introduction

Cone beam computed tomography (CBCT) is a rapidly developing technology that is used both for imaging in radiotherapy and for sectional imaging during interventional procedures. Raising the awareness of healthcare staff about radiation doses to patients from CBCT and other medical procedures they are delivering or prescribing is an ongoing challenge. Values for some dose related quantities are displayed on CT imaging equipment. For example the volume weighted CT dose index ($CTDI_{vol}$) is displayed routinely on multi-slice CT (MSCT) scanners and the weighted CT dose index ($CTDI_w$) is displayed on Varian cone beam CT (CBCT) scanners known as the On-Board Imager (OBI) system linked to radiotherapy equipment. These quantities perform an essential role in reporting the radiation level to allow comparison of dose levels with measured values and facilitate optimization, but they appear to clinicians as abstract quantities that are only vaguely related to the doses received by patients. The $CTDI_w$ and $CTDI_{vol}$ represent the mean doses deposited in standard phantoms made of polymethyl methacrylate (PMMA) known as CTDI phantoms, but were derived originally to provide a dose similar to that received by the tissues within the section of the patient's body that was scanned. Radiotherapy clinicians are familiar with absorbed doses from treatment to tissues surrounding the target volume, and the availability of information indicative of organ doses from imaging procedures used for image guided radiation therapy (IGRT) would provide them with a more comprehensive picture of doses from the whole treatment programme. Although the standard phantoms in which dose quantities are measured have cross sections and compositions that are substantially different from those of the human body, the CTDI has the potential to provide an indication of organ dose levels (Turner *et al* 2010). Therefore, it should be possible to establish links between the measured dose quantities and organ doses, and this would improve the knowledge and understanding of radiotherapists and other radiation professionals in healthcare. The quantity displayed on the OBI system is the $CTDI_w$ for a reference beam of width 20 mm, resembling that for conventional MSCT.

A study has been made of CBCT scans employed in the Beatson West of Scotland Cancer Centre for IGRT procedures. The study has used Monte Carlo (MC) simulations on the International Commission on Radiological Protection ICRP-110 adult male and female reference computational phantoms (ICRP 2009) to evaluate mean organ and tissue absorbed doses. The investigation aimed to assess whether CTDI values for the reference beam width ($CTDI_{w,ref}$) displayed on the OBI system could be used to give clinicians indications of doses to specific organs irradiated during IGRT procedures in order to help inform their risk/benefit judgements.

2. Materials and methods

2.1. $CTDI_{100}$ for CBCT dosimetry

$CTDI_{100}$ is based on normalizing an integrated dose profile from a single axial rotation at the centre of a scan ($z = 0$) with respect to the nominal beam width used for the scan. Since CBCT scans are acquired using a single rotation of a wide cone, a reference beam width (W) equal to 20 mm is used to evaluate the $CTDI_{100}$ for dosimetry purposes:

$$\text{CTDI}_{100} = \frac{1}{W} \int_{-50 \text{ mm}}^{50 \text{ mm}} D(z) dz \quad (1)$$

CTDI_{100} measurements used to evaluate the weighted CTDI (CTDI_w) displayed on the CBCT console are made within cylindrical PMMA phantoms that are 160 mm or 320 mm in diameter, and 150 mm in length, and are designed to emulate a reference adult head or body. The CTDI_{100} measurements are made at the middle of the central axis ($\text{CTDI}_{100,c}$) and in four peripheral positions ($\text{CTDI}_{100,p}$) in the phantoms. The CTDI_w which takes account of the dose distribution over the x - y plane is evaluated from the two measurements using the equation:

$$\text{CTDI}_w = \frac{1}{3} \text{CTDI}_{100,c} + \frac{2}{3} \text{CTDI}_{100,p} \quad (2)$$

where $\text{CTDI}_{100,p}$ is the average of four measurements at the peripheral axes ($\text{CTDI}_{100,p} = 1/4 \sum_{i=1}^4 \text{CTDI}_{100,p,i}$). The $\text{CTDI}_{100,c}$ and $\text{CTDI}_{100,p}$ are measured with the beam collimated to give a beam width of 20–22 mm. The precise position of the collimator plates is monitored electronically and appropriate adjustments made to correct for small differences from the 20 mm prescribed. Such measurements are carried out after maintenance checks and upgrades to confirm the accuracy of the displayed $\text{CTDI}_{w,\text{ref}}$.

2.2. CBCT scanning protocols and CTDI_w measurements

The kV imaging system employed in this study to acquire CBCT scans was the OBI system with a software version of (V1.6). The system is incorporated into a Varian TrueBeam linear accelerator (Varian Medical systems, Palo Alto, CA) and utilized in the clinic in IGRT procedures for patient positioning prior to radiotherapy treatment. The OBI system has been described in detail in previous studies (Abuhaimeed *et al* 2014, 2015). Three pre-configured scanning protocols, head, thorax, and pelvis are commonly employed in IGRT procedures, and have been used for this study. Parameters of these protocols are listed in table 1, together with values for $\text{CTDI}_{w,\text{ref}}$ acquired with a reference beam of width 20 mm within the standard PMMA phantoms and displayed on the scanner.

2.3. Monte Carlo simulations and dose calibration

MC simulation is widely used for assessing organ doses from CT, CBCT, and other diagnostic radiology procedures. Simulations based on different MC codes have been utilized to estimate organ doses from CBCT scans employed in IGRT procedures (Ding and Munro 2013, McMillan *et al* 2013, Brochu *et al* 2014, Nelson and Ding 2014, Poirier *et al* 2014, Son *et al* 2014). MC user codes BEAMnrc (Rogers *et al* 1995) and DOSXYZnrc (Walters *et al* 2013) based on EGSnrc code (V4-r2-4-0, National Research Council of Canada, Ottawa, Canada) (Kawrakow 2000) were used to simulate the OBI system and to calculate doses to organs and tissues in three dimensions (3D) in terms of absorbed dose. The MC model designed with BEAMnrc has been benchmarked against various experimental measurements and good agreement obtained (Abuhaimeed *et al* 2014, 2015). Organ doses were assessed by using the MC codes in two stages:

2.3.1. Calibration of Monte Carlo model. The DOSXYZnrc code, which was used to assess organ doses, reports absorbed dose values in (Gy/incident particle). Therefore, in order to evaluate the absorbed dose for each organ and tissue in Gy, the MC code was calibrated by

Table 1. Parameters employed in the CBCT scanning protocols for the head, thorax, and pelvis.

Scan parameter	Head	Thorax	Pelvis
X-ray voltage (kV)	100	125	125
Exposures (mAs)	147	264	1056
Projections number	367	660	660
Acquisition mode	Full-Fan	Half-Fan	Half-Fan
Gantry rotation (degrees)	200° (90°–290°) ^a	360°	360°
Bowtie filter	Full	Half	Half
Longitudinal extent of beam (W) (mm)	198	198	198
Scan diameter (mm)	264	478	478
CTDI _{w,ref} (mGy)	2.85	3.48	13.94

^a 90° is left of the patient.

using the method described in Qiu *et al* (2012). This method is based on obtaining a MC calibration factor (F_{MC}) as:

$$F_{MC} = \frac{D_{Exp}}{D_{MC} \times \text{mAs}} \quad (3)$$

where D_{MC} is absorbed dose calculated in DOSXYZnrc (Gy/incident particle), and D_{Exp}/mAs is absorbed dose measured experimentally in (Gy/mAs). Thus, F_{MC} is defined in terms of (incident particles/mAs). Experimental measurements for D_{Exp} were made in terms of absorbed dose to water at a depth of 2 cm as recommended in the Institution of Physics and Engineering in Medicine code of practice (Klevenhagen *et al* 1996, Aukett *et al* 2005). A depth of 2 cm is sufficient to avoid the surface region where there is greater variation in dose with depth, and curves depicting depth dose normalized with respect to the dose at 2 cm are reproducible. While use of a greater depth would give a lower signal for measurement.

$$D_{w,z=2} = MN_k k_{ch} \left[\left(\frac{\mu_{en}}{\rho} \right)_{w/air} \right]_{z=2, \varphi} \quad (4)$$

where $D_{w,z=2}$ is absorbed dose to water at a depth of 2 cm, M is the chamber response in coulombs (C) corrected for air temperature and pressure, N_k is the air kerma calibration factor for the chamber employed in (Gy/C), k_{ch} is a correction factor which accounts for the difference between the chamber response in air and inside the water phantom, and $\left[\left(\frac{\mu_{en}}{\rho} \right)_{w/air} \right]_{z=2, \varphi}$

is the mass energy absorption coefficient ratio water to air averaged over the photon spectrum at a depth of 2 cm within the water phantom using a field of diameter φ . Values for D_{Exp} were measured at a depth of 2 cm within a $30 \times 30 \times 20 \text{ cm}^3$ solid water phantom (Gammex, Middleton, WI, US) set up at a source-to-surface distance (SSD) of 100 cm using a Farmer ionization chamber NE2611A calibrated at the UK National Physical Laboratory (NPL) for energies in the kV range. The measurements were performed with a stationary kV source using a field of size $10 \times 10 \text{ cm}^2$ at a SSD of 100 cm. Two formats were used to cover the scanning protocols studied (table 1): (1) Head: 100 kV and full bowtie filter, and (2) Body: 125 kV and half bowtie filter. The measurements were repeated three times using 1000 mAs/reading for each protocol, and the average reading M was subsequently normalized with the respect to the mAs used for the measurements to obtain (C/mAs). N_k values were provided in the calibration

certificate for the beam qualities specified in terms of half value layer (HVL), whereas values for k_{ch} and $\left[\left(\frac{\mu_{\text{en}}}{\rho}\right)_{\text{w/air}}\right]_{z=2, \varphi}$ were obtained from the code of practice (Klevenhagen *et al* 1996).

The same parameters and conditions, under which D_{Exp} values were measured, were used in the MC calculations to derive D_{MC} values. First, the validated BEAMnrc model was used to run two simulations for the two protocols used for D_{Exp} with 1×10^9 histories using ISOURCE: 10 of the code (Parallel Circular Beam Incident from Side). The variance reduction technique called the directional bremsstrahlung splitting (DBS) with the splitting number (NBRSP) set to 2×10^4 was used to increase the efficiency of the simulations (Kawrakow *et al* 2004, Mainegra-Hing and Kawrakow 2006). The cut-off energies for transport of photons (PCUT) and the low energy thresholds for creation of secondary photons (AP) were set to 1 keV, whereas energies of electrons were deposited locally at the interaction site (Ding and Munro 2013). The bremsstrahlung cross sections, bremsstrahlung angular sampling, and photon cross sections were based on the data of the National Institute of Standards and Technology (NIST), Koch–Motz, and NIST XCOM, respectively (Mainegra-Hing and Kawrakow 2006). The simulations were run using the default boundary crossing algorithm (EXACT) and electron step algorithm (PRESTA-II). Spin effect, the electron impact ionization, Rayleigh scattering, and atomic relaxations were turned on (Ding and Coffey 2010). Outputs of the simulations were recorded in phase space (PHSP) files at a SSD of 100 cm.

The resulting PHSP files were then employed as sources in DOSXYZnrc using ISOURCE: 2 of the code (Phase-Space Source Incident from Any Direction). The solid water phantom used for D_{Exp} measurements was designed in DOSXYZnrc. 3×10^8 histories with the photon splitting number set to 100 were run to calculate D_{MC} values at a depth of 2 cm with a statistical uncertainty of $\leq 0.2\%$. The same MC parameters and the cross sectional data used in BEAMnrc simulations were employed for DOSXYZnrc simulations.

Once D_{Exp} and D_{MC} had been determined, values of F_{MC} at a depth of 2 cm for the head and body protocols were calculated for 1 mAs as in equation (3). Subsequently, F_{MC} was utilized to evaluate absolute absorbed doses (Gy) within the organs and tissues for each scanning protocol using the mAs values listed in table 1 as:

$$D_{\text{Exp}} = D_{\text{MC}} \times F_{\text{MC}} \times \text{mAs} \quad (5)$$

2.3.2. Calculation of organ doses. After calibration of the MC code, two simulations were run in BEAMnrc using fields of size $264 \times 198 \text{ mm}^2$ for the head scan and $478 \times 198 \text{ mm}^2$ for the thorax and pelvis scans. The simulations were run with the same MC parameters and numbers of histories employed for the calibration in section 2.3.1. The PHSP files were recorded at a SSD of 75 cm, and these were used subsequently in DOSXYZnrc as kV sources with a distance set to 25 cm to assess organ and tissue doses at a source-isocentre-distance (SID) of 100 cm, which represented typical values in clinical practice.

Organ doses were evaluated in DOSXYZnrc using the ICRP-110 adult male and female reference computational phantoms, which were based on whole body CT scans acquired for real patients (ICRP 2009). The main characteristics of the phantoms are given in table 2. Each phantom is provided by the ICRP in a 3D matrix with sizes of $254 \times 127 \times 222$ and $299 \times 137 \times 348$ for the male and female phantoms, respectively. Each cell of the two matrices contains a unique ID number between 1 and 141, which represents a pre-defined organ or tissue. Each ID number is assigned to one of 53 different media, which were defined with different elemental compositions and densities to simulate the organs and tissues. DOSXYZnrc format, however, also takes the form of a 3D matrix, but it consists of voxels of

Table 2. The main characteristics of the ICRP-110 male and female reference computational phantoms used in this study (ICRP 2009).

Property	Male	Female
Age (year)	38	43
Height (m)	1.76	1.63
Mass (kg)	73.0	60.0
Slice thickness (mm)	8.0	4.84
Voxel in-plane resolution (mm)	2.137	1.775
Voxel volume (mm ³)	36.54	15.25
Number of columns	254	299
Number of rows	127	137
Number of slices	222	348

specific sizes, and each voxel is assigned to a specific medium. Thus, the first step required converting the ICRP-110 phantoms to DOSXYZnrc format. A MATLAB-based code was developed, and the phantoms were converted using the same voxel size as the ICRP CT scans $2.137 \times 2.137 \times 8.0 \text{ mm}^3$ for the male and $1.775 \times 1.775 \times 4.84 \text{ mm}^3$ for the female. The mediums assigned for the DOSXYZnrc voxels were based on the elemental compositions and densities defined in the ICRP-110 report (ICRP 2009). All the media used in DOSXYZnrc were defined by using PEGS4 code. Figure 1 shows the coronal and sagittal views of the phantoms converted to DOSXYZnrc format, which were in agreement with those given in the ICRP-110 report (ICRP 2009).

The centre of the head scan ($z = 0$) was positioned at the middle of the head phantom, and the axial centres of the thorax and pelvic scans positioned at the middle of the lungs and pelvis, respectively. The numbers of histories run for the male and female phantoms were 4×10^9 and 8×10^9 , respectively, in order to obtain statistical uncertainties of $<1\%$ for voxels of all organs and tissues located within the scan field. ISOURCE: 8 of the code (Phase-Space Source Incident from Multiple Directions) was employed with projection numbers and gantry rotation set to the values listed in table 1. The MC parameters used for the calibration were also employed in evaluation of the organ doses.

In order to evaluate the absorbed dose for each organ and tissue in 3D efficiently, a MATLAB-based code was developed to analyse the outputs of the DOSXYZnrc simulations, which are given in (.3ddose) format. As each organ and tissue was defined with an ID number, three 3D matrixes were created for each simulation. The first matrix was based on the original ICRP file, where each cell contained the ID number. The second and third matrixes were based on the 3ddose file, where each voxel of the second matrix contained an absorbed dose value, and a statistical uncertainty value was recorded in the third matrix. These matrixes were linked together in the MATLAB code, thus the first step was to specify coordinates of the voxels that carried the same ID number in the first array. The second step was to obtain the mean absorbed dose D_{MC} and the mean statistical uncertainty values for each ID number from the second and third matrixes based on the coordinates specified in the first step.

The mean absorbed dose for each organ and tissue is calculated from either single or multiple ID numbers as described in the ICRP-110 report (ICRP 2009). For example, the dose for the brain is represented by the ID number 61, while that for the salivary glands is calculated from ID numbers 120 and 121, which represent the left and right salivary glands, respectively. For organs and tissues evaluated from a single ID number, the mean absorbed dose was calculated as the mean value over the all voxels assigned. However, for organs and tissues with multiple ID numbers, the mean absorbed dose was calculated as the weighted value, as each ID number had different voxel numbers, using:

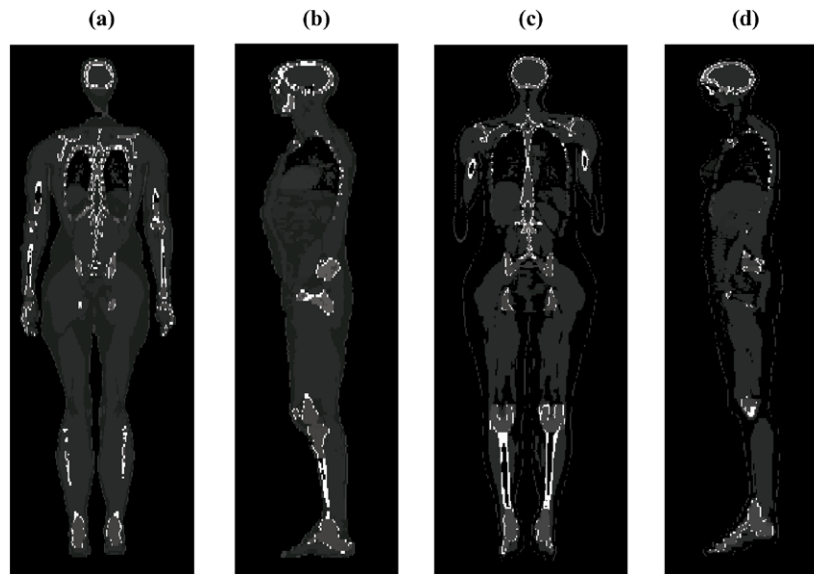


Figure 1. The coronal and sagittal views of the ICRP-110 (a) and (b) male and (c) and (d) female reference computational phantoms converted to DOSXYZnrc format.

$$D_{MC} = \frac{\sum_{i=1}^n D_{MC,i} VN_i}{\sum_{i=1}^n VN_i} \quad (6)$$

where n is number of the ID numbers used to calculate the mean dose for a given organ or tissue, and $D_{MC,i}$ and VN_i are the mean absorbed dose and voxel numbers for each ID number i , respectively.

The MATLAB code used to analyse the 3ddose files was validated by comparing doses for a selection of organs against known values calculated manually and by using STATDOSE code (McGowan *et al* 2013). Organs were chosen that were more sensitive to radiation and lay within the field of the relevant scan. Doses for brain from head scan, and prostate, rectum, and bladder from pelvic scan were also used to allow comparison with data from a study by Montanari *et al* (2014).

The percentage differences between the $CTDI_{w,ref}$ and the calculated mean absorbed doses to organs and tissues (OD) were calculated as:

$$\text{Difference} = \left(\frac{OD}{CTDI_{w,ref}} - 1 \right) \times 100\% \quad (7)$$

3. Results and discussion

3.1. Organ doses from CBCT scans

The mean absorbed doses to tissues and organs located within the primary beam for each CBCT scan calculated using the MC code for the ICRP reference phantoms are listed in tables 3–5, together with the doses per 100 mAs to allow comparison with other data. The

Table 3. Calculated means and ranges of absorbed doses within organs and tissues of the ICRP-110 reference male and female phantoms and absorbed doses per 100 mAs from the head scan protocol factors listed in table 1.

Organ or tissue	Male organ doses (mGy)			Female organ doses (mGy)		
	Mean tissue dose	Range of dose in tissue	Dose per 100 mAs	Mean tissue dose	Range of dose in tissue	Dose per 100 mAs
Bone-marrow (red)	3.8	1.2–6.4	2.6	5.4	1.7–9.1	3.7
Extrathoracic (ET) region	2.3	1.6–3.0	1.6	3.6	2.8–4.4	2.5
Brain	3.0	1.4–4.7	2.1	3.4	1.4–5.4	2.3
Salivary glands	4.5	3.9–5.1	3.1	5.3	4.8–5.9	3.7

Table 4. Calculated means and ranges of absorbed doses within organs and tissues of the ICRP-110 reference male and female phantoms and absorbed doses per 100 mAs from the thorax scan protocol factors listed in table 1.

Organ or tissue	Male organ doses (mGy)			Female organ doses (mGy)		
	Mean tissue dose	Range of dose in tissue	Dose per 100 mAs	Mean tissue dose	Range of dose in tissue	Dose per 100 mAs
Bone-marrow (red)	6.7	4.5–8.8	2.5	8.1	6.0–10.2	3.1
Lung	7.6	6.3–8.9	2.9	9.7	8.4–11.0	3.7
Stomach	3.4	0.9–6.0	1.3	2.7	0.1–5.5	1.0
Breast	7.5	7.4–7.7	2.9	9.9	9.4–10.5	3.8
Oesophagus	6.1	3.5–8.8	2.3	7.9	4.5–11.3	3.0
Liver	3.2	1.1–5.3	1.2	4.0	0.7–7.3	1.5
Thyroid	2.5	1.8–3.2	1.0	3.3	2.0–4.6	1.3
Heart	10.0	8.3–11.7	3.8	12.6	11.0–14.1	4.8
Pancreas	1.2	0.9–1.5	0.4	0.9	0.6–1.2	0.3
Spleen	3.4	1.9–4.9	1.3	3.4	1.6–5.2	1.3
Thymus	11.2	8.8–13.6	4.2	13.2	12.4–14.0	5.0
Gall bladder	1.2	0.8–1.7	0.5	1.5	1.2–1.9	0.6

statistical uncertainties in the dose calculations within each voxel for the majority of organs and tissues were between 0.1% and 0.6%. The range of doses within different parts of each tissue, shown in tables 3–5, give an indication of the variation in dose throughout the tissue. The ranges were smaller for organs and tissues that lay fully within the scanned region, and large for ones that were only partially in the beam or more distributed. Although the scanning protocols were the same for the male and female phantoms, organ doses tended to be greater for the female phantom. The disparities arose from differences between the phantoms in height, which resulted in a proportionately longer section of the abdominal organs being irradiated in the female, and diameter, so that the overall attenuation of the beam in tissue overlying the sensitive organs is less in the female (table 2). These differences are larger than would be expected with MSCT, because the same field size and scan parameters of CBCT are used routinely for imaging all patients, whereas in MSCT the section of the body imaged would be adapted based on the scan projection radiograph.

In head scan, doses for the female phantom brain and salivary glands were 10% and 15% higher, respectively, than those for the male, while doses for other tissues were 30% to 40% higher in the female phantom (table 3). The doses to organs in the head appear to be affected

Table 5. Calculated means and ranges of absorbed doses within organs and tissues of the ICRP-110 reference male and female phantoms and absorbed doses per 100 mAs from the pelvic scan protocol factors listed in table 1.

Organ or tissue	Male organ doses (mGy)			Female organ doses (mGy)		
	Mean tissue dose	Range of dose in tissue	Dose per 100 mAs	Mean tissue dose	Range of dose in tissue	Dose per 100 mAs
Bone-marrow (red)	29.1	22.3–35.8	2.8	34.0	28.3–39.7	3.2
Colon	10.7	8.6–12.8	1.0	26.9	20.8–33.0	2.5
Prostate/Uterus	32.4	30.1–34.6	3.1	36.6	33.5–39.7	3.5
Small intestine	14.3	0.7–29.4	1.4	24.2	4.4–44.1	2.3
Gonads (Testes/Ovary)	3.4	2.8–4.0	0.3	35.8	34.7–37.0	3.4
Urinary bladder	41.6	32.7–50.6	3.9	46.4	36.0–56.8	4.4
Kidneys	1.3	1.0–1.6	0.1	2.3	1.8–2.9	0.2
Rectum	36.6	35.7–37.5	3.5	37.6	35.8–39.5	3.6

Table 6. A comparison between organ doses in (mGy/100 mAs) calculated in this study using ICRP-110 adult male phantom and those from (Montanari *et al* 2014).

Organ or tissue	This study	Montanari <i>et al</i> (2014)
Head scan		
Brain	2.1 (0.9–3.2)	1.9 (1.3–2.3)
Pelvic scan		
Prostate	3.1 (2.9–3.3)	3.0 (1.7–4.5)
Rectum	3.5 (3.4–3.5)	3.2 (2.1–4.2)
Bladder	3.9 (3.1–4.8)	3.1 (1.7–4.6)

by the different angulations of the head in the male and female phantoms. Doses to organs and tissues in the male and female thorax scans were more similar, all being within 30% of each other (table 4). Absorbed doses to organs in the female phantom for the pelvic scan, were all higher than in the male phantom (table 5). Those for the bladder and rectum in the two phantoms were within 10%, but doses for the colon, small intestine and kidneys differed by a factor of two or more. The colon dose in the female was more than double that in the male, partly because the primary beam was more attenuated in the thicker male phantom and partly because a larger proportion of the colon lay within the scan field for the shorter female. Similar findings with respect to the colon have been reported in a study using the ICRP 110 reference phantom conducted with MSCT (Zhang *et al* 2012) and for organs in the head in dental CBCT scans (Morant *et al* 2013, Zhang *et al* 2013).

Results for some organs and tissues in terms of dose per 100 mAs are compared with those from a study by Montanari *et al* (2014) in table 6. Their study used MC simulations on CT scans for 25 adult brain and 25 prostate cancer patients to calculate organ doses resulting from head and pelvic CBCT scans using a similar OBI system, and doses for the two assessments are within 10%–20% of each other.

3.2. Comparison of the $CTDI_{w,ref}$ with mean absorbed organ and tissue doses

In figures 2–4, values for the $CTDI_{w,ref}$ are compared with doses from tables 3–5 for a selection of the more radiosensitive organs in the reference phantoms for the three examination protocols listed in table 1. For the purpose of the comparisons, the organ doses were normalized

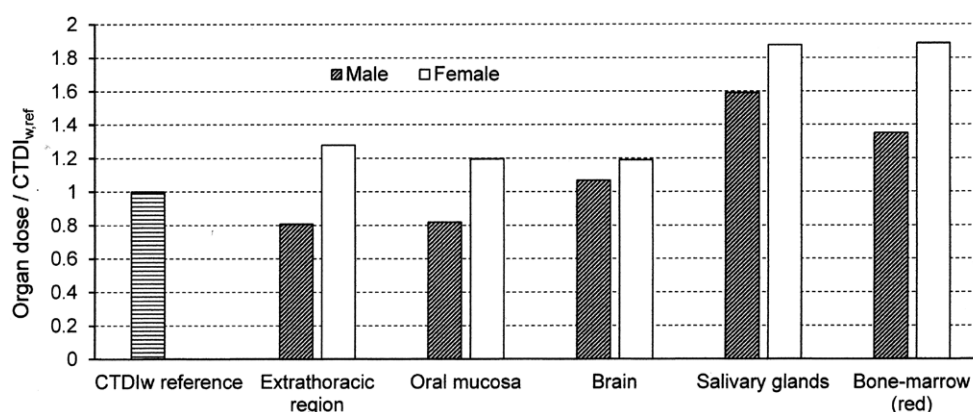


Figure 2. Comparison of values for the $CTDI_{w,ref}$ (horizontal shading) with mean absorbed doses to a selection of tissues in the ICRP male and female reference phantoms. The absorbed doses were derived by MC simulation for scans of the head and normalized with respect to the $CTDI_{w,ref}$.

with respect to the $CTDI_{w,ref}$ derived for a 20mm wide reference beam. For the head scan, mean absorbed doses in the brain were 10–20% higher than the $CTDI_{w,ref}$ for the male and female phantoms, but doses to the salivary glands were substantially greater (figure 2). Absorbed doses in the extrathoracic region and oral mucosa were about 20% higher than the $CTDI_{w,ref}$ in the female phantom and 20% lower in the male.

For the thorax scans, values for the stomach and liver were within $\pm 21\%$ of the $CTDI_{w,ref}$, which could therefore provide a potential indicator for doses to organs in the lower thorax (figure 3). In contrast, absorbed doses to the lung, breast and oesophagus were double the $CTDI_{w,ref}$ or even greater. Doses to the lung tissue are likely to be higher because of the lower attenuation and the difference in shape of the anatomical and measurement phantoms. Doses to the breast and oesophagus are higher because they are more superficial. Since the agreement between male and female organ doses appears reasonable for the thorax scan, a factor of two could be applied to the $CTDI_{w,ref}$ and this used as an estimate of doses to these organs for both phantoms. The dose to the thyroid was within 27% of the $CTDI_{w,ref}$ for both phantoms, but since the thyroid is near the upper boundary of the scan and in the neck, which has a substantially smaller diameter, the $CTDI_{w,ref}$ is unlikely to provide a reliable dose assessment.

Absorbed doses for organs and tissues irradiated in the pelvic scans were more variable (figure 4). The dose to the male colon was lower than the $CTDI_{w,ref}$, and that to the small intestine was similar, while doses to the corresponding female organs were almost double the $CTDI_{w,ref}$. Doses to the bladder, prostate, uterus and gonads were 2–3 times the $CTDI_{w,ref}$. Doses to the red bone marrow are substantially larger than those to soft tissue organs in all the scans (figures 2–4), despite the small area of the body irradiated. This is because of the higher absorption coefficient of bone. In the majority of the scans, the mean absorbed dose to the red bone marrow was about double the value of the $CTDI_{w,ref}$.

3.3. Use of the $CTDI_{w,ref}$ to estimate organ doses

The $CTDI_{w,ref}$ that is displayed by the OBI system is comparable to the mean absorbed doses to a number of organs and tissues in the reference phantoms that lie within the scan field. It could provide an assessment that is sufficient to give operators and clinicians an indication of the mean absorbed doses delivered to a few designated organs in the ICRP reference phantoms

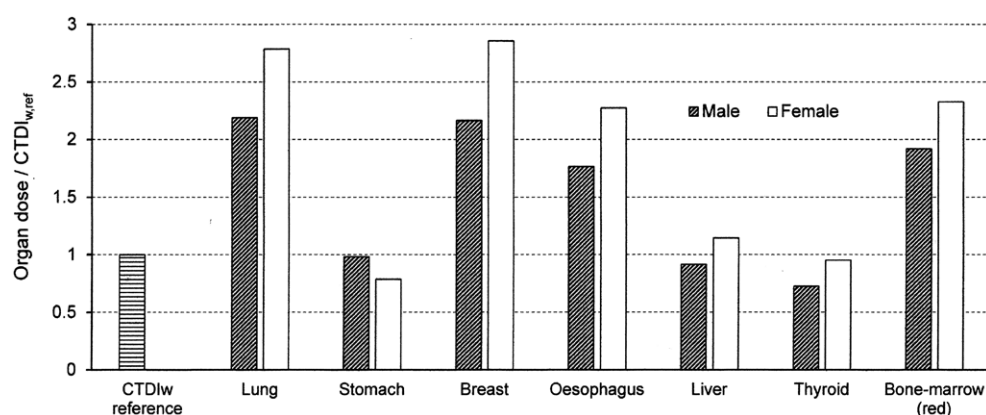


Figure 3. Comparison of values for the $CTDI_{w,ref}$ (horizontal shading) with mean absorbed doses to a selection of tissues in the ICRP male and female reference phantoms. The absorbed doses were derived by MC simulation for scans of the thorax and normalized with respect to the $CTDI_{w,ref}$.

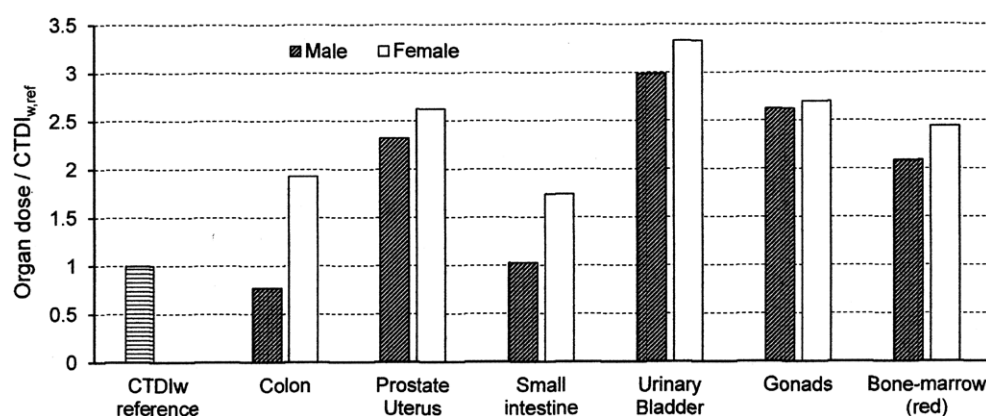


Figure 4. Comparison of values for the $CTDI_{w,ref}$ (horizontal shading) with mean absorbed doses to a selection of tissues in the ICRP male and female reference phantoms. The absorbed doses were derived by MC simulation for scans of the pelvis and normalized with respect to the $CTDI_{w,ref}$.

by imaging procedures that they can relate to the doses delivered by the therapy exposures. Percentage differences in mean absorbed doses for a selection of organs or tissues from the $CTDI_{w,ref}$ calculated using equation (7) are listed in table 7. The $CTDI_{w,ref}$ underestimated doses to the lungs, female breast, oesophagus and bone marrow significantly, but reasons for this are apparent and doubling the index would give values close to the organ doses. Thus, multiplying the $CTDI_{w,ref}$ by two provides a possible option for deriving an assessment for some organs that are important in determining the overall risk and this adjustment is included in table 7. In principle these results should be applicable to other CBCT systems used for image guided radiotherapy (IGRT) with which a similar approach to imaging is used.

The uncertainties in any assessment of organ and tissue doses based on simulations involving reference phantoms are large by their very nature, as they do not take account of large differences between individuals in body size and shape. Components of the uncertainties in

Table 7. Difference between the mean absorbed dose and $\text{CTDI}_{\text{w,ref}}$ factor for a selection of organs and tissues in the ICRP-110 reference male and female phantoms.

Organ or tissue	Male	Female
Head scan		
Brain	+7	+19
Bone marrow ($\text{CTDI}_{\text{w,ref}} \times 2$)	−33	−6
Thorax scan		
Oesophagus ($\text{CTDI}_{\text{w,ref}} \times 2$)	−12	+14
Breast ($\text{CTDI}_{\text{w,ref}} \times 2$)	+8	+43
Lung ($\text{CTDI}_{\text{w,ref}} \times 2$)	+9	+39
Stomach	−2	−21
Liver	−8	+15
Bone marrow ($\text{CTDI}_{\text{w,ref}} \times 2$)	−4	+16
Pelvic scan		
Colon	−23	+93
Bone marrow ($\text{CTDI}_{\text{w,ref}} \times 2$)	+4	+22

Note: Differences were calculated as in equation (7).

organ and tissue doses within a reference phantom are the measurement and calibration of the incident radiation ($\pm 20\%$), tissue densities and compositions ($\pm 10\%$), boundaries of the scanned region of the body ($\pm 10\%$), and variations in shape and geometry of the organs ($\pm 15\%$ – 40%) (Martin 2007). The exact positions of the boundaries of the cone beam will determine the proportions of the organs lying near the edge that are irradiated. Doses to these organs can vary by 30–40% or more, so it is not possible to estimate doses to these tissues using a simple index. Thus, accurate evaluation of absorbed doses in organs even in relation to a reference patient is difficult to achieve at this time.

The differences in organ doses between the reference male and female phantoms confirm these large variations. The differences between the doses for the two phantoms are larger than might be expected for a MSCT scanner, because the exposure factors and the size of the radiotherapy CBCT scan field are not adapted for individual patients routinely, so greater proportions of some organs will be exposed for smaller patients. Monte Carlo simulations of doses to radiosensitive organs that take into account patient size, indicate that the ratio of organ dose to CTDI_{vol} decreases with patient size, because of attenuation in the overlying layers of adipose tissue (Turner *et al* 2010, Li *et al* 2011, Huda and He 2012). Therefore, if the same exposure factors are used, the doses to radiosensitive organs and tissues of larger patients will be lower.

In principle, less radiation should be required to image smaller patients, because of the smaller height and lower attenuation. However, the exposure factors and collimation are not adapted routinely for individual patients. A standard field is used to ensure precise positioning of the target tissue for delivery of the therapy dose. Raising the awareness of doses received from CBCT imaging should help to promote the implementation of optimization through the use of lower exposure factors for smaller patients and possibly encourage improved beam collimation based on an initial radiographic image. However, this must always take into account the main aim of CBCT scans, which is to assist the delivery of a large dose to the treatment site (tumour) with a high degree of accuracy.

It is apparent from the analyses performed (figures 2–4 and table 7) that the $\text{CTDI}_{\text{w,ref}}$ could give approximate values for absorbed doses to organs within the scan field for reference phantoms, which represent reference patients, but not for organs lying in peripheral

positions. $\text{CTDI}_{\text{w,ref}}$ values are within about $\pm 20\%$ of the absorbed doses for the brain in the head scan, and the stomach and liver in the thorax scan. An option to improve accuracy for individual patients would be to apply correction factors based on body water equivalent diameter to the $\text{CTDI}_{\text{w,ref}}$, as is done with the size specific dose estimate (SSDE) for MSCT (AAPM 2011, 2014). It is useful to have the information at the time of the scan, without the need for additional calculation, so incorporation of suitable software in CBCT scanners to provide a quantity similar to the SSDE based on body dimensions derived from the scan images is recommended as a future development for CBCT dosimetry. Nevertheless, the values of $\text{CTDI}_{\text{w,ref}}$ displayed currently can provide radiotherapy clinicians with an indication of the doses that are being delivered by imaging to other organs in the region of the body being treated. Raising the awareness of clinicians of the link between $\text{CTDI}_{\text{w,ref}}$ and organ dose is recommended, so that they can factor the knowledge of the exposure level into plans for their patients. The assessment could be improved further in the future if volumes of irradiated tissues were determined from individual scans (DeMarco *et al* 2007) and such software should become available in time as processing capabilities improve.

4. Conclusions

The purpose of the CTDI_{w} is to provide a practical measure that can be used to evaluate scanner performance and compare different scanners under a reference condition. It provides a link to the dose that is used by physicists and radiographers as a quality assurance index. Doses to radiosensitive organs and tissues determine the risk from radiation, and it would be helpful to have more information about these doses to improve the knowledge and understanding of clinical staff in radiotherapy. This study has shown that the $\text{CTDI}_{\text{w,ref}}$ for a narrow reference beam on a CBCT scanner can provide an estimate of the absorbed doses to organs and tissues within the scan field that is sufficient to give a general indication of dose levels from imaging. For thorax scans, the $\text{CTDI}_{\text{w,ref}}$ can be used directly as a measure of doses to the stomach and liver, and $2 \times \text{CTDI}_{\text{w,ref}}$ can be used as a measure of dose to the lung, breast and oesophagus. For head scans the $\text{CTDI}_{\text{w,ref}}$ can give a measure of brain dose, and for pelvic scans, it can provide an indication of colon dose, although differences between results for male and female phantoms give an indication of the approximate nature of the link. For all the scans $2 \times \text{CTDI}_{\text{w,ref}}$ can give a measure of the likely absorbed dose to the red bone marrow. It is proposed that knowledge of this link should be included in the training of radiotherapy staff to increase their awareness of doses from CBCT imaging.

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